



GAME DEVELOPMENT ROADMAP

Step-by-Step Guide to Become a Game Developer



LEARN



PRACTICE



BUILD



POLISH



PUBLISH



GROW

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1. Fundamentals

- Understand how games work
- Game development workflow
- Learn programming basics (C# recommended)
- Mathematics for games (Vectors, Trigonometry)

2-4 Weeks

Resources
Brackeys, FreeCodeCamp, Codecademy

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2. Learn a Game Engine

- Choose a engine: Unity (Beginner Friendly)
- Unity / Unreal Engine basics
- Editor interface, Scenes, GameObjects
- Components, Prefabs, Scripts

3-4 Weeks

Resources
Unity Learn, Brackeys, Unreal Online Learning

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3. Core Concepts

- Transform, Camera, Lighting
- Physics, Colliders, Rigidbodies
- Input handling
- UI Basics

3-4 Weeks

Resources
Unity Learn, Documentation, YouTube Tutorials

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4. Gameplay Programming

- Player Controller
- Enemies & AI Basics
- Health, Damage, Scoring System
- Game States & Managers

4-6 Weeks

Resources
Brackeys, GameDev.tv, YouTube

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5. Assets & Level Design

- Importing Assets
- Materials, Textures, Shaders
- Level / Environment Design
- Prototyping your ideas

3-4 Weeks

Resources
Unity Asset Store, Quixel, Kenney Assets

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6. Audio & Visual Effects

- Sound & Music Integration
- Particle Systems
- Animations & Transitions
- Post Processing

3-4 Weeks

Resources
Unity Learn, FMOD, Mixamo

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7. Data & Game Systems

- Save / Load System
- Inventory System
- Dialog System
- Scriptable Objects

3-4 Weeks

Resources
Unity Docs, GameDev.tv, GitHub Examples

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8. Advanced Topics

- AI Advanced (Pathfinding, Behavior Trees)
- Multiplayer Basics (Photon / Mirror)
- Optimization (Performance, Memory)
- Design Patterns

4-6 Weeks

Resources
Unity Docs, GDC Talks, GameDev.tv

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9. Testing & Debugging

- Debugging Techniques
- Performance Profiling
- Playtesting & Iteration
- Bug Fixing Best Practices

2-3 Weeks

Resources
Unity Profiler Docs, Gamasutra

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10. Build & Publish

- Build Settings
- Platform Optimization (PC, Mobile, Web)
- Publishing to Steam / Play Store
- Monetization Basics

3-4 Weeks

Resources
Unity Docs, Steamworks, Google Play Docs

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11. Portfolio & Projects

- Complete 3-5 Projects
- Showcase on Portfolio Website
- GitHub for Code Projects
- Get Feedback & Improve

Ongoing

Resources
Personal Projects, Itch.io, GitHub

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12. Build Your Career

- Apply for Game Dev Jobs / Internships
- Contribute to Game Jams
- Network in Game Dev Communities
- Keep Learning & Stay Updated

Ongoing

Resources
LinkedIn, Game Jams, Discord Communities

ESSENTIAL SKILLS

- Programming (C# / C++)
- Problem Solving & Logic
- Game Design Principles
- 3D Math & Physics
- Creativity & Innovation
- Attention to Detail
- Teamwork & Communication
- Persistence & Learning

TOOLS & TECHNOLOGIES

- | | |
|---------------|--------------------|
| Unity | C# |
| Unreal Engine | Visual Studio Code |
| Visual Studio | Aseprite |
| Blender | Audacity / FMOD |
| Git & GitHub | Trello / Notion |
| Photoshop | Firebase |

PROJECT IDEAS (PRACTICE)

- ✓ 2D Platformer Game
- ✓ Top-Down Adventure Game
- ✓ 3D Runner Game
- ✓ Space Shooter Game
- ✓ Puzzle Game
- ✓ Multiplayer FPS (Basic)
- ✓ Mobile Hyper Casual Game

LEARNING PATH FLOW



CAREER OPPORTUNITIES

- | | |
|---------------------|----------------------|
| Game Developer | Technical Artist |
| Gameplay Programmer | QA Tester |
| Game Designer | Game Producer |
| Level Designer | Indie Game Developer |



CONSISTENCY + PRACTICE + CREATIVITY = GAME DEVELOPMENT SUCCESS

